

# BENJAMIN HOFFMAN

[bhoffman@ethz.ch](mailto:bhoffman@ethz.ch) | [LinkedIn](#) | [Personal Website](#) | Swiss & American

## EDUCATION

### ETH ZURICH

#### PhD Researcher in the Networked Systems Group

- ▶ Current research: machine learning, control, and algorithms for large scale systems and networks
- ▶ Supervised by Prof. Laurent Vanbever and affiliated PhD researcher at the ETH AI Center

Zurich, Switzerland

Feb. 2025 - present

### ETH ZURICH

#### MSc in Robotics, Systems and Control — GPA: 5.7/6.0

- ▶ Focus on control and machine learning

Zurich, Switzerland

Sep. 2022 - Dec. 2024

### LUND UNIVERSITY

#### Academic Exchange Semester — GPA: 5.7/6.0

- ▶ Focus on control and machine learning

Lund, Sweden

Jan. 2022 - Jun. 2022

### ETH ZURICH

#### BSc in Mechanical Engineering — GPA: 5.2/6.0

- ▶ Focus on mechatronics and control

Zurich, Switzerland

Sep. 2018 - Aug. 2021

### KANTONSSCHULE WETTINGEN

#### Swiss Matura and IB Diploma — GPA: 5.7/6.0 (top 0.5%, salutatorian)

- ▶ Major in law and economics, academic exchange semester at the Gymnase de Nyon (VD, Switzerland)

Wettingen, Switzerland

Aug. 2014 - Jun. 2018

## RESEARCH EXPERIENCE

### NETWORKED SYSTEMS GROUP — ETH ZURICH

#### Student Research Assistant — part time

- ▶ Implemented a scalable real-world data collection and algorithm evaluation platform using Python and Amazon MTurk
- ▶ Performed distributed testing of ML-based streaming algorithms to evaluate their performance in real-world conditions

Zurich, Switzerland

May 2024 - Dec. 2024

### MASTER THESIS — ETH AI CENTER | LEARNING & ADAPTIVE SYSTEMS GROUP

#### Sample Efficient Model-Based Reinforcement Learning — Grade: 5.75/6.0 ([view here](#))

- ▶ Using model-based RL for efficient policy learning on a Boston Dynamics Spot quadruped with JAX, PyTorch, and C++
- ▶ Learning robot dynamics from data by combining Bayesian Neural Networks with physical priors
- ▶ Supervised by Jin Cheng, Chenhao Li, Bhavya Sukhija, Lenart Treven, Prof. Stelian Coros, and Prof. Andreas Krause

Zurich, Switzerland

Jun. 2024 - Dec. 2024

### SEMESTER THESIS — NETWORKED SYSTEMS GROUP

#### Machine Learning Based Video Streaming Algorithms — Grade: 6.0/6.0 ([view here](#))

- ▶ Researched and evaluated state-of-the-art ML-based video streaming algorithms
- ▶ Developed Python-based network-data collection and benchmarking platform using Docker and AWS/Azure SDKs
- ▶ Supervised by Dr. Alexander Dietmüller and Prof. Laurent Vanbever

Zurich, Switzerland

Sep. 2023 - Jan. 2024

### BACHELOR THESIS — COMPUTATIONAL ROBOTICS LAB

#### State Estimation for Legged Robots — Grade: 6.0/6.0 ([view here](#))

- ▶ Researched, developed, and implemented different state estimation approaches with C++ and a Unitree A1 quadruped
- ▶ Determined the optimal algorithm still being used in the Lab to date based on an analysis of speed and accuracy
- ▶ Supervised by Dr. Dongho Kang and Prof. Stelian Coros

Zurich, Switzerland

Feb. 2021 - Aug. 2021

## PUBLICATIONS

- [1] **Benjamin Hoffman**, Alexander Dietmüller, Ayush Mishra, and Laurent Vanbever. “Strategic Data Collection and Generalization for Learning-Based Adaptive Bitrate Control”. In: *Under Review* (2025).
- [2] **Benjamin Hoffman**, Alexander Dietmüller, Ayush Mishra, and Laurent Vanbever. “Into the Wild: Real-World Testing for ML-Based ABR”. In: *ACM PACMI* (Seoul, South Korea, 2025). [[PDF](#)]
- [3] **Benjamin Hoffman**, Alexander Dietmüller, and Laurent Vanbever. “Poster: Towards Network Model Generalization using Strategic Data Collection”. In: *ACM SIGCOMM* (Coimbra, Portugal, 2025). [[PDF](#)]
- [4] **Benjamin Hoffman**, Jin Cheng, Chenhao Li, and Stelian Coros. “Learning More With Less: Sample-Efficient Model-Based RL for Loco-Manipulation”. In: *Workshop on Resource-Rational Robot Learning at CoRL* (Seoul, South Korea, 2025). [[PDF](#)]

WORK EXPERIENCE

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>BAIN &amp; COMPANY</b><br><b>Associate Consultant Intern — full time intern (return offer received)</b><br>‣ Advised 3 industry leading firms and Fortune 500 companies on performance improvement and IT strategy projects<br>‣ Conducted empirical research and analyzed large, unstructured datasets to formulate strategic recommendations               | <b>Zurich, Switzerland</b><br>Feb. 2024 - Apr. 2024 |
| <b>ERANEOS</b><br><b>Data Science Intern — full time intern (return offer received)</b><br>‣ Worked on 5+ projects in finance and the public sector with a focus on ML & GenAI, LLMs, and data analytics<br>‣ Developed custom LLM solutions and multi-agent systems using fine-tuned local LLMs, RAG, and OpenAI API                                           | <b>Zurich, Switzerland</b><br>Jun. 2023 - Dec. 2023 |
| <b>GÜDEL GROUP</b><br><b>Robotics and Software Engineer — full time intern and part time working student (return offer received)</b><br>‣ Machine learning, data science, and software engineering for robotics and industrial analytics<br>‣ Transformed 3+ quality control procedures to ML-driven processes and rewrote the software for 4+ key applications | <b>Bern, Switzerland</b><br>Aug. 2021 - May 2023    |
| <b>ETH ZURICH</b><br><b>Teaching Assistant for Analysis III and Dimensioning I — part time</b><br>‣ Prepared and led weekly exercise sessions for 30+ students, graded assignments, and exam supervision<br>‣ Elected by students to lead a two-day exam preparation course with 30+ participants for Analysis III                                              | <b>Zurich, Switzerland</b><br>Sep. 2020 - Jan. 2023 |

EXTRACURRICULAR EXPERIENCE

|                                                                                                                                                                                                                                                                                                                      |                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>UN-ETH STUDENT TEAM</b><br><b>Project Lead</b><br>‣ Building a voting-data processing pipeline and a user-facing visualization interface for all 3 main UN bodies<br>‣ In charge of project management, leading a team of 5 software engineers, and coordinating with UN stakeholders                             | <b>Zurich, Switzerland</b><br>Sep. 2025 - present         |
| <b>GLP SIGGENTHAL</b><br><b>Board and Committee Member</b><br>‣ In charge of communications, as well as planning and implementing campaigns and initiatives on a local level<br>‣ Member of the advisory body of our municipality on topics of nature and environmental conservation                                 | <b>Obersiggenthal, Switzerland</b><br>Aug. 2021 - present |
| <b>VEREIN LEGAL HELP (NGO)</b><br><b>Board Member and Head of IT</b><br>‣ In charge of strategic initiatives, maintaining long-term growth, as well as all technical and software related projects<br>‣ Led an organization-wide (100+ users) software transition project resulting in savings of 5000+ CHF annually | <b>Zurich, Switzerland</b><br>Sep. 2022 - Sep. 2025       |
| <b>ANALYTICS CLUB AT ETH</b><br><b>Quants at ETH Member</b><br>‣ Growing the quant community at ETH, organizing events of 70+ participants at ETH with leading quant firms                                                                                                                                           | <b>Zurich, Switzerland</b><br>Feb. 2023 - Dec. 2024       |
| <b>ETH ENTREPRENEUR CLUB</b><br><b>Finance Committee Member</b><br>‣ Led a project to automate reimbursements, invoicing, and accounting, increasing process efficiency by over 300%<br>‣ Acted as the principal financial point of contact for leading startups, 50+ members, and major corporate partners          | <b>Zurich, Switzerland</b><br>Sep. 2020 - Feb. 2022       |

SKILLS & CERTIFICATIONS

|                       |                                                                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PROGRAMMING</b>    | Python, C++, MATLAB, JavaScript, SQL                                                                                                                              |
| <b>FRAMEWORKS</b>     | PyTorch, JAX, Docker, Pandas, NumPy, SciPy, scikit-learn, OpenAI Gym & API, LangChain, Git, AWS, Azure<br>NVIDIA Isaac Sim, ROS, Boston Dynamics, Unitree, OpenCV |
| <b>LANGUAGES</b>      | English (Native), German (Native), Swiss German (Native), French (B2)                                                                                             |
| <b>CERTIFICATIONS</b> | J&S Coach Tennis & Snowboard, Certificate in Portfolio and Risk Management (University of Geneva, 2023)                                                           |

INTERESTS

|                        |                                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>SPORTS</b>          | Competitive tennis player (coach and team captain), snowboarding, running, strong interest in health and fitness |
| <b>OTHER INTERESTS</b> | Public policy, law and economics, global and local politics, nature and the environment                          |